

Sibabrata Biswal

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Professional Experience

Quarry Business Management

Ongoing

- Manage quarry business operations and developed an **ERP system** to digitize and streamline business workflows, operations, and management.

AI Automation & Business Integration, Freelance

Ongoing

- Design and implement **AI automations and business integrations** for SMEs, improving workflows through LLMs, APIs, and software automation.

Jawaharlal Nehru University, Research Associate

New Delhi, IN

Prof. B. R. Panda, School of Biotechnology [Website](#)

Feb 2025 – Aug 2025

AI-based Prediction of Metastasis in Tongue Cancer

- Fine-tuned **deep learning models** on radiology data for tumor classification/localization; generated **Grad-CAM** and attention maps.

PreprintCommons – Scientific Preprint Discovery

- Built LLM-based metadata extraction pipeline and full-stack platform using **React, FastAPI, PostgreSQL, TailwindCSS**.

Deepgene, C-CAMP, Research Assistant

Bangalore, IN

- Worked on **PreprintCommons** and cancer diagnosis projects under Prof. B. R. Panda; continued project work at JNU.

Jigyasu, DASCO Education Pvt. Ltd., Academic Manager [Website](#)

Bhubaneswar, IN

- Managed **academic operations, sessions, and content/product development** for digital learning programs.
- Conducted **client technical demonstrations** and contributed to digital branding and academic technology initiatives.

Research Projects

Deep Learning-based Modeling for Predicting Drug-Kinase Bioactivity [Master's Thesis]

Dr. V. Badireenath Konkimalla, NISER [Website](#)

- Developed **GNN drug-kinase interaction predictor** using ChEMBL, combining **RDKit descriptors + ProtBert embeddings**; MAE 1.02, RMSE 1.28.
- Built **PyTorch + DGL** encoder-decoder pipeline with stratified graph splitting for drug-repurposing prediction.

TiH-IoT Foundation, IIT Bombay, Project Fellow

Predictive Irrigation Using IoT & Machine Learning

- Developed IoT plant-monitoring and ML pipeline for predictive irrigation; contributed to sensor integration and experimental design.

Publications

Book Chapter

Predictive irrigation: current practice and future prospects

Subhrajyoti Mishra, **Sibabrata Biswal**, Anuleho Biswas, Abhijit Chakraborty, Subhankar Mishra

DOI: [10.1016/B978-0-443-24139-0.00022-9](https://doi.org/10.1016/B978-0-443-24139-0.00022-9)

Conference Proceeding

IoT4Irrigation – Integrating IoT and Machine Learning for Low-cost Sustainable Agriculture

Sibabrata Biswal, et al.

DOI: [10.1007/978-3-031-74701-4](https://doi.org/10.1007/978-3-031-74701-4)

Education

Integrated Master of Sciences | Life Sciences

2019 – 2024

National Institute of Science Education and Research (NISER), Bhubaneswar

CGPA: 7.55/10.0

Applied Generative AI & AI Engineering

2026

100x Engineers – Applied GenAI Cohort

Technical Skills

Programming:

Python, R, JavaScript, SQL, Bash, C

AI/ML:

PyTorch, DGL, Scikit-learn, LLMs, RAG, Agents, MCP

Bioinformatics:

RDKit, PyMOL, GROMACS, AutoDock/Vina

Computational Biology:

RNA-Seq, WGBS, MNase-Seq, WGCNA, Cytoscape

Web/Tools:

React, FastAPI, PostgreSQL, TailwindCSS, Git, Linux, LaTeX